



DATABASE SYSTEM • PROJECT REVIEW

Alumni Mentoring & Engagement

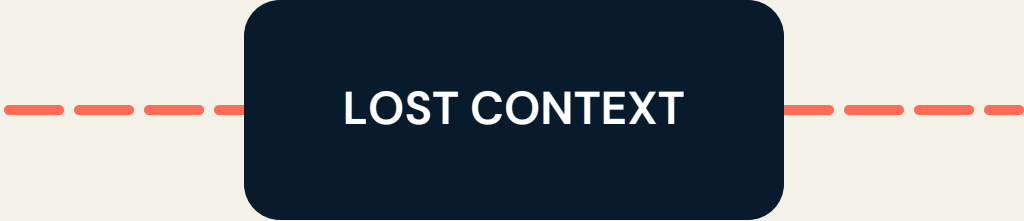
Design and Implementation of a Database Management System

4 SEPTEMBER 2026 • PRESENTER NAME

Engagement exists. Institutional memory does not.

TODAY

Social groups
Spreadsheets
Scattered messages



LOST CONTEXT

NEEDED

One reliable history

Profiles, mentoring, events,
donations and feedback managed
as connected records.

One system. Five connected domains.

ALUMNI
PROFILES

MENTORING
PROGRAMMES

EVENT
MANAGEMENT

DONATIONS

FEEDBACK

OBJECTIVES

Efficient management • Data integrity • Actionable reporting • User-friendly access

The scope follows the full engagement lifecycle—from first profile to long-term institutional insight.

Four roles around one trusted record

ALUMNI

Profiles • mentoring • giving

STUDENTS

Guidance • session requests

ENGAGEMENT
DATABASE

```
graph TD; A[ALUMNI  
Profiles • mentoring • giving] --- E[ENGAGEMENT DATABASE]; S[STUDENTS  
Guidance • session requests] --- E; AS[ADMIN STAFF  
Governance • coordination] --- E; M[MENTORS  
Expertise • availability] --- E;
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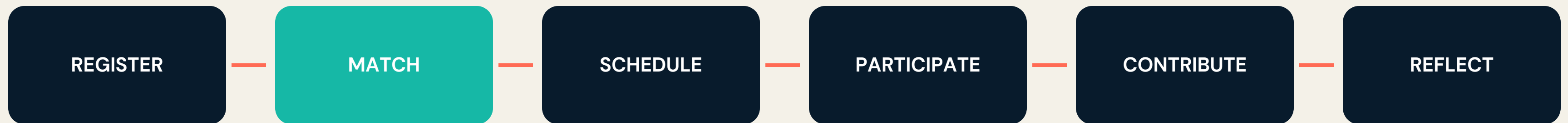
ADMIN STAFF

Governance • coordination

MENTORS

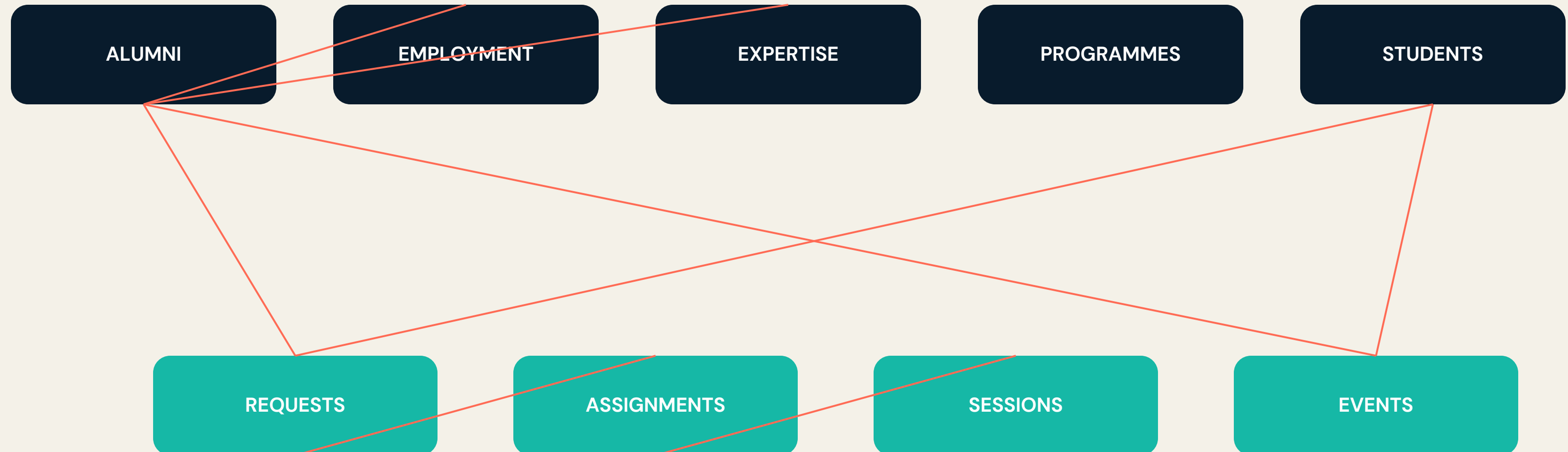
Expertise • availability

A traceable journey from registration to insight



Profile management → mentoring requests and matching → session tracking → event attendance → donation reporting → feedback collection

Conceptual ER design



Donations and Feedback connect to Alumni, Events and completed Sessions.

The relational schema makes every link explicit

ALUMNI
PK alumni_id
UQ email
programme_id FK

MENTORING_REQUEST
PK request_id
student_id FK
expertise_id FK

ASSIGNMENT
PK assignment_id
request_id FK
mentor_id FK

SESSION
PK session_id
assignment_id FK
scheduled_at

EVENT_REGISTRATION
PK registration_id
event_id FK
alumni_id FK

DONATION
PK donation_id
alumni_id FK
amount

Normalization turns raw records into dependable structure

UNNORMALIZED

Repeating groups

1NF

Atomic values

2NF

No partial dependency

3NF

No transitive dependency

Business rules enforced at the data boundary



ONE ACTIVE MENTOR
per request



VALID SESSION DATE
future scheduling only



EVENT CAPACITY
never exceeded



UNIQUE EMAIL
per user



POSITIVE DONATION
amount > 0